

Best Practice for Bulk RNA-seq analysis

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- Library prep for RNA-seq
- Upstream Analysis Workflow
- Downstream Analysis Workflow
- From BAM to Count
- Differential Gene Expression
- Enrichment of DEGs with Over Representation Analysis (ORA)
- RNA-seq databases: GTEx and TCGA

Aim of the methods

Bulk DNA-seq?

Bulk RNA-seq?



Aim of the methods

Bulk DNA-seq

4.2 Mapping reads to reference

Reference: ...TCGAATGCG...	
Read1:	CTCGAATACG
Read2:	CTCGAATACG
Read3:	CTCGAATACG
Read4:	CTCGAATACG
Read5:	GCGAATACG
Read6:	GCGAATACG
Read7:	GCGACTACG
Read8:	GCGAATACG

Maternal

Paternal

Error

Heterozygous

Homozygous

4.3 Calling variants

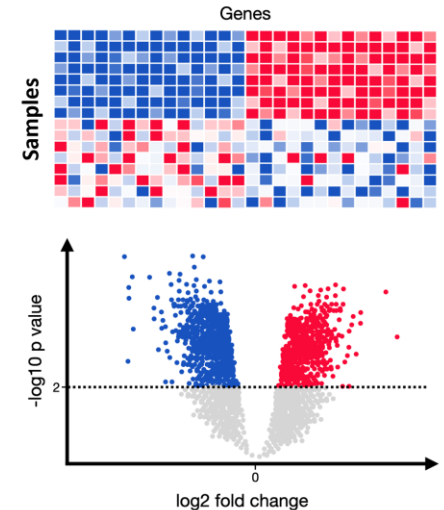
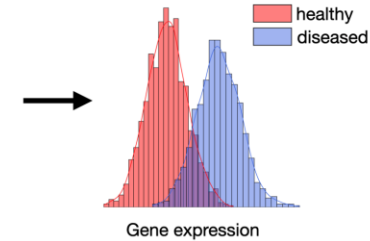
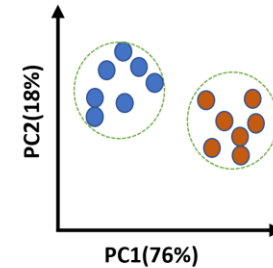
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##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">								
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">								
##FORMAT=<ID=AD,Number=2,Type=Integer,Description="Read depth for each allele">								
#CHROM	POS	ID	REF	ALT	QUAL	FILTER	FORMAT	Sample1
20	14370	rs6054257	T	G	129	PASS	GT:GQ:DP:AD	0/1:48:8:4,4
20	17330	.	G	A	150	PASS	GT:GQ:DP:AD	1/1:49:8:8,8

ANN=G|stop_gained|HIGH|OR4F5|ENSG00000186092|transcript|ENST0000641515.2|protein_coding|3/3|c.822T>G|p.Trp274*|882/2618|822/981|274/326||Pathogenic

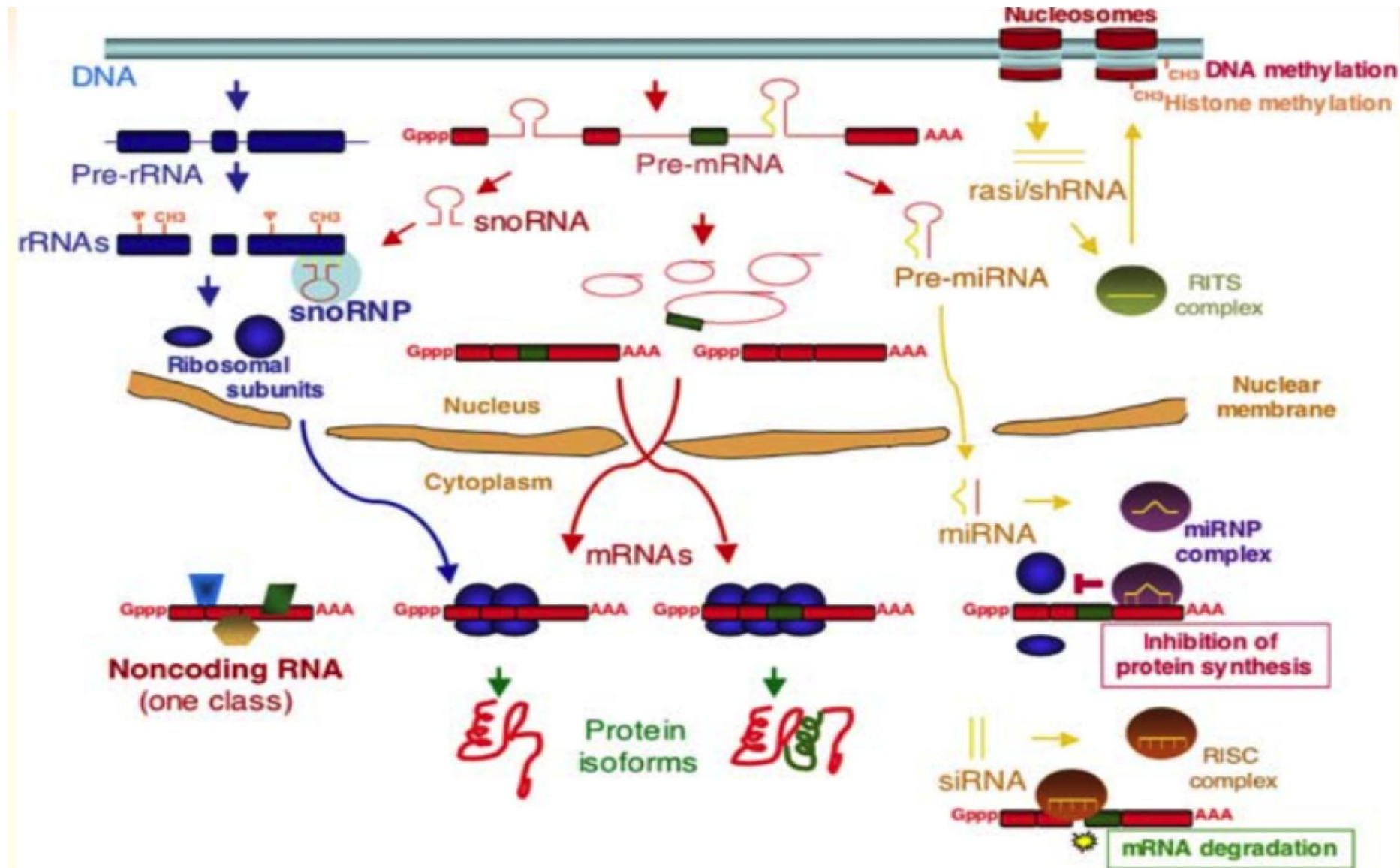
ANN=A|frameshift_variant|HIGH|ZSWIM2|ENSG00000163012|transcript|ENST00000295131.3|protein_coding|9/9|c.1238G>A|p.Ile413|1293/2451|1238/1902|413/633||LOF=(ZSWIM2|ENSG00000163012|1|1.00)

4.4 Annotating variants

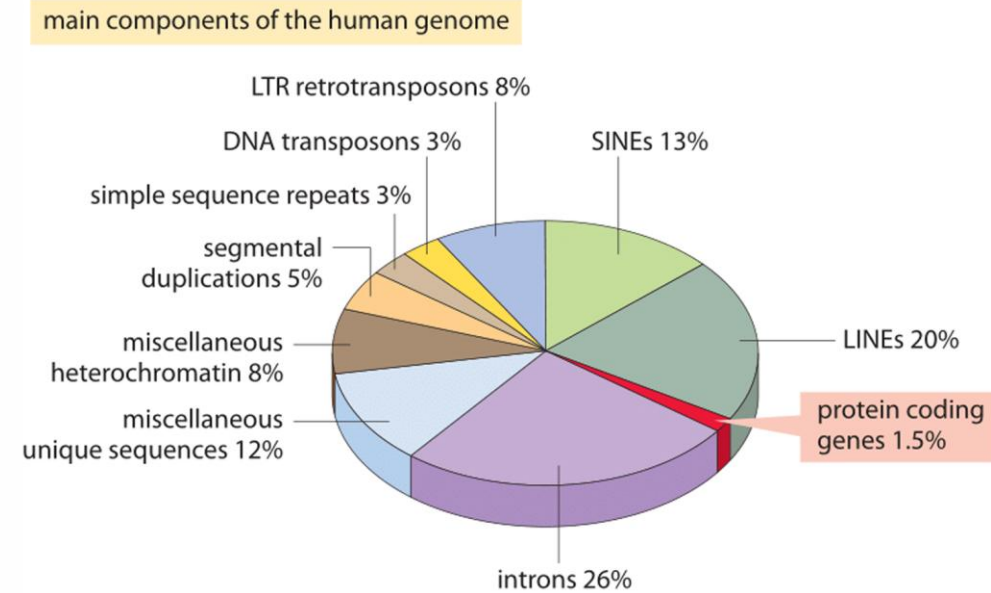
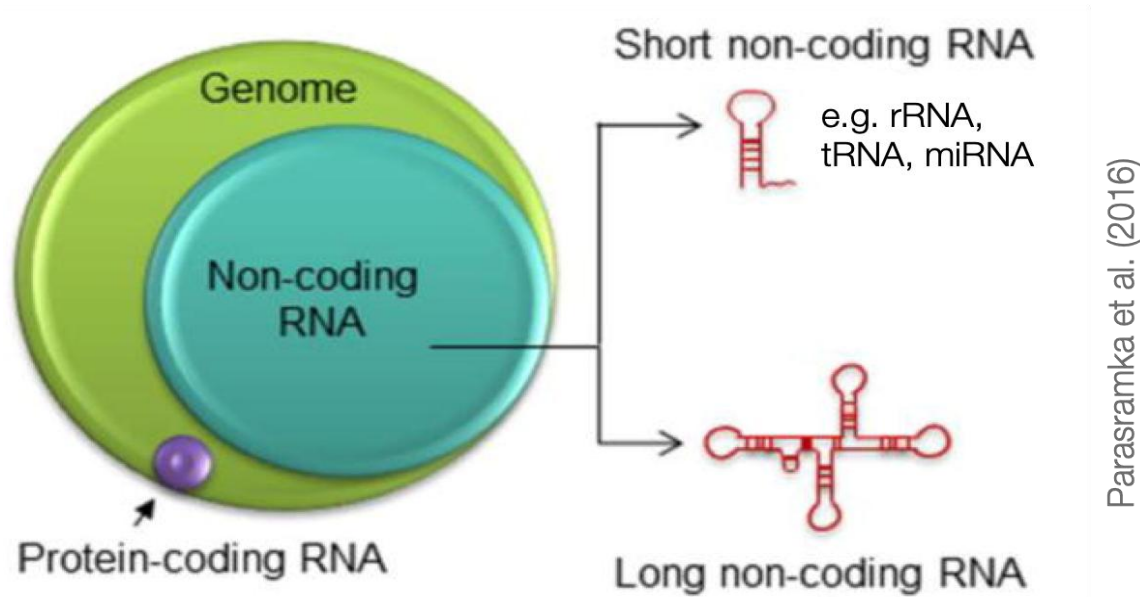
Bulk RNA-seq



Different types of RNA



Different types of RNA

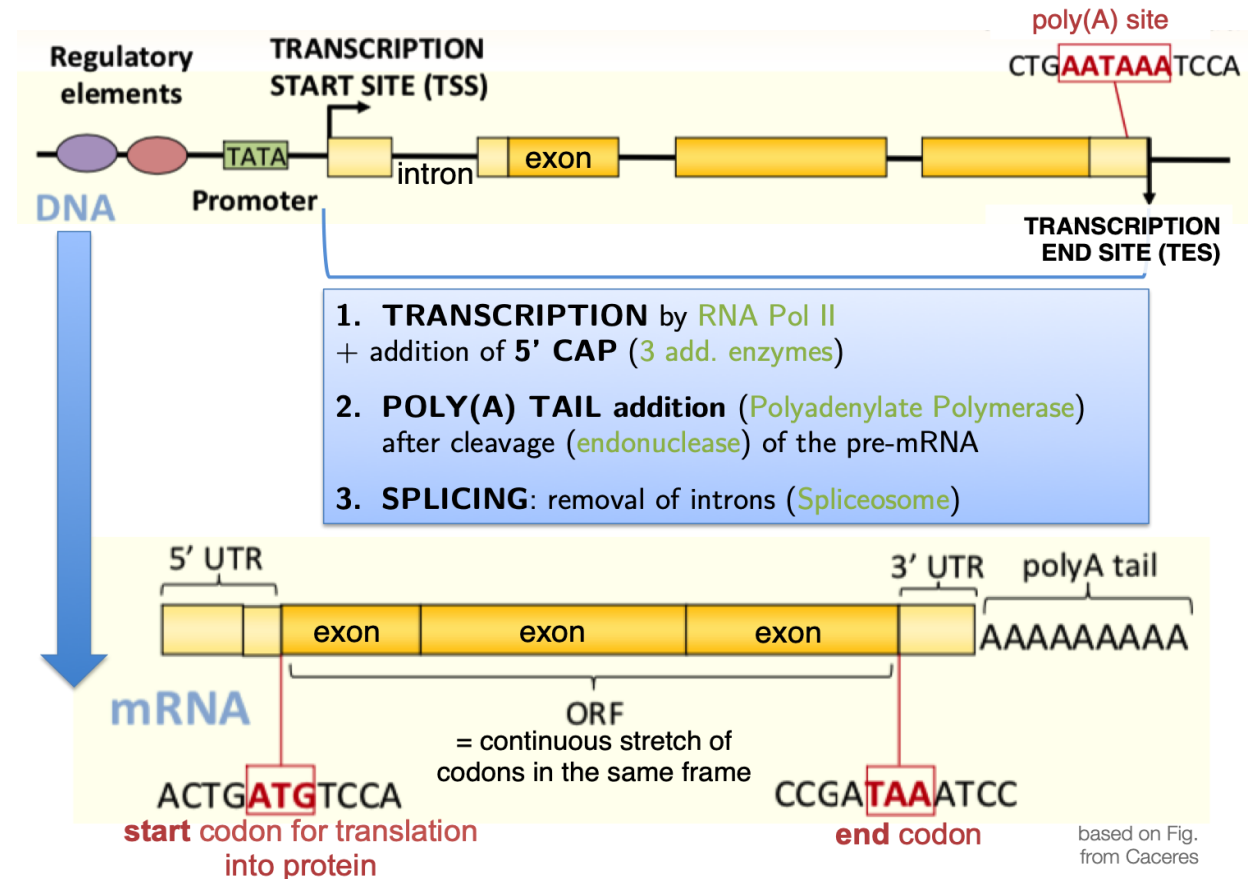


» How many genes are in a genome? (bionumbers.org)

Sequencing library prep protocol depends on the RNA properties

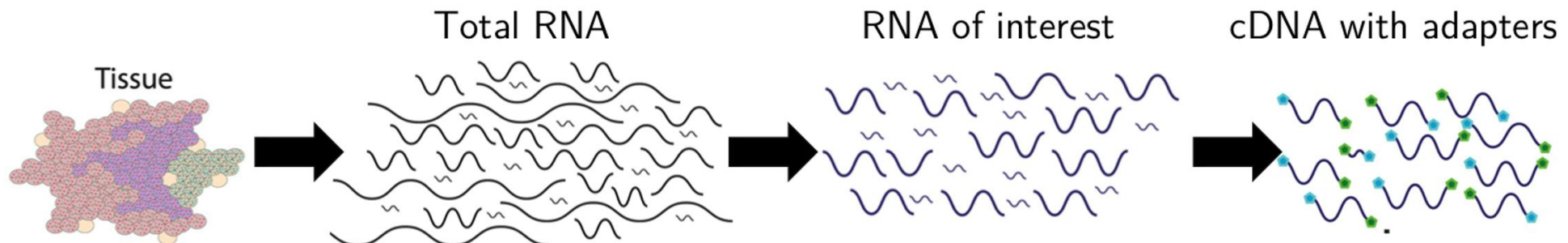
It is not a one-size-fits-all situation!

- Abundance and stability
 - ▶ rRNA: 90-95% (!)
 - ▶ tRNA: 3-5%
 - ▶ mRNA: 2%
 - ▶ all other non-coding RNAs: *well* below 1%
- Cellular location
 - ▶ most are in the cytoplasm
- Size
 - ▶ miRNAs: 18-23bp
 - ▶ mRNA: several 100 to 1000 bp
- Specific sequences/modifications
 - ▶ poly(A) tails of mRNA
 - ▶ 2D structure
 - ▶ antisense transcripts



General steps of RNA-seq preparation

- 1 RNA extraction (cell lysis, RNA purification)
- 2 **enrichment of the RNA of interest**
 - ▶ mRNA: poly(A) enrichment vs. ribosomal-depletion
 - ▶ small RNAs: size-based enrichment
- 3 fragmentation (ca. 200 bp)
- 4 cDNA synthesis
- 5 library prep to obtain cDNA with adapters for sequencing



General steps of RNA-seq preparation

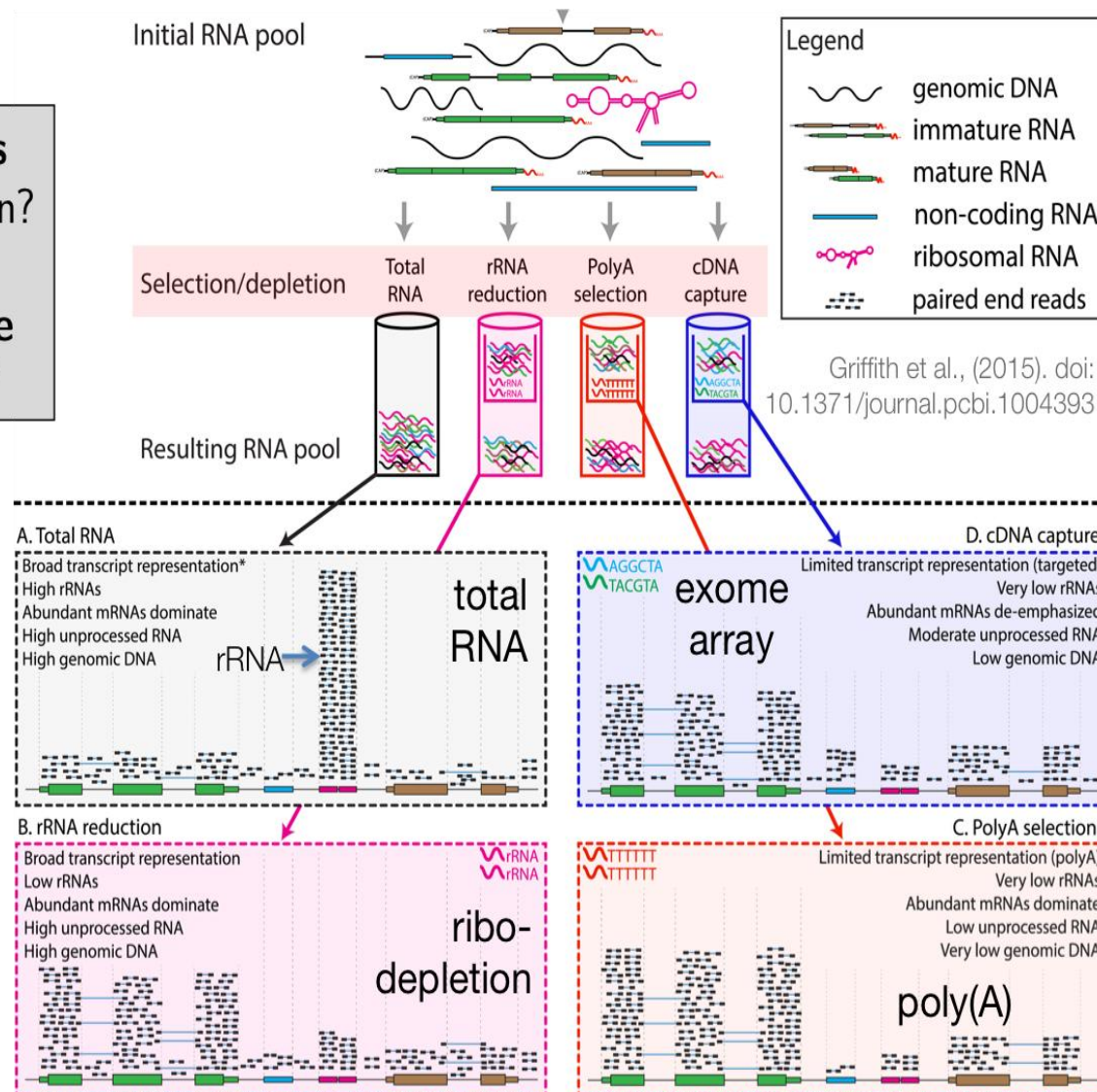
which **transcripts**
are you interested in?

what type of **noise**
can you tolerate?

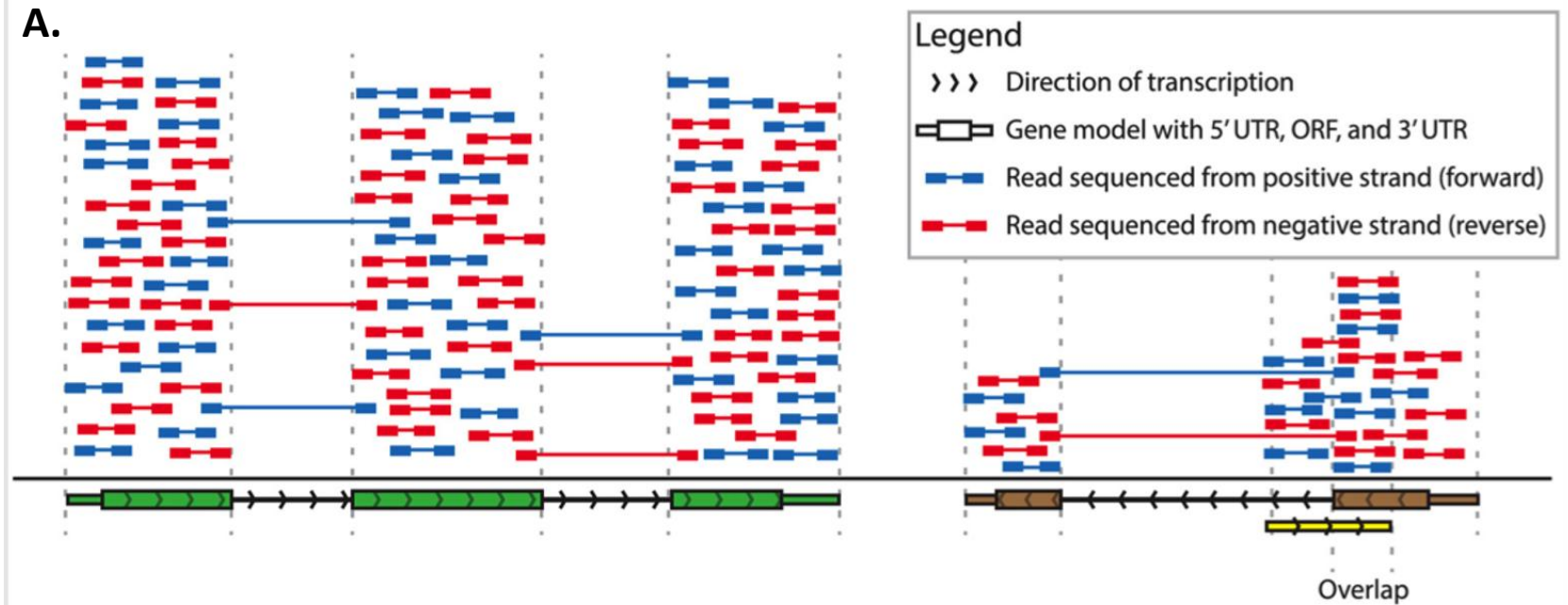
■ rRNA

■ protein coding
(strongly expressed)

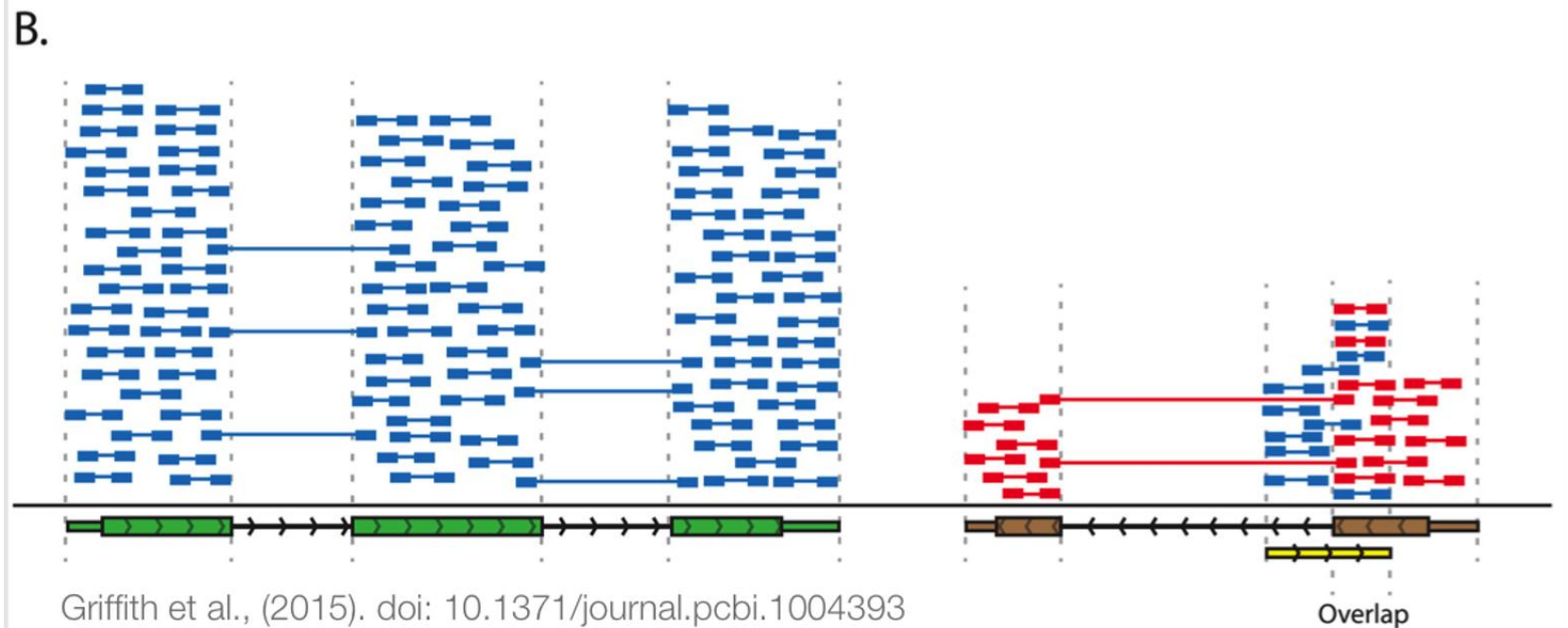
■ protein coding
(lowly expressed)



Non-
stranded

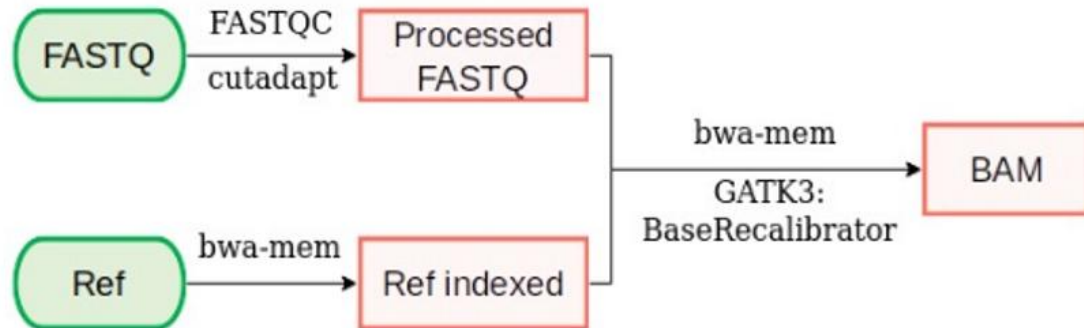


Stranded

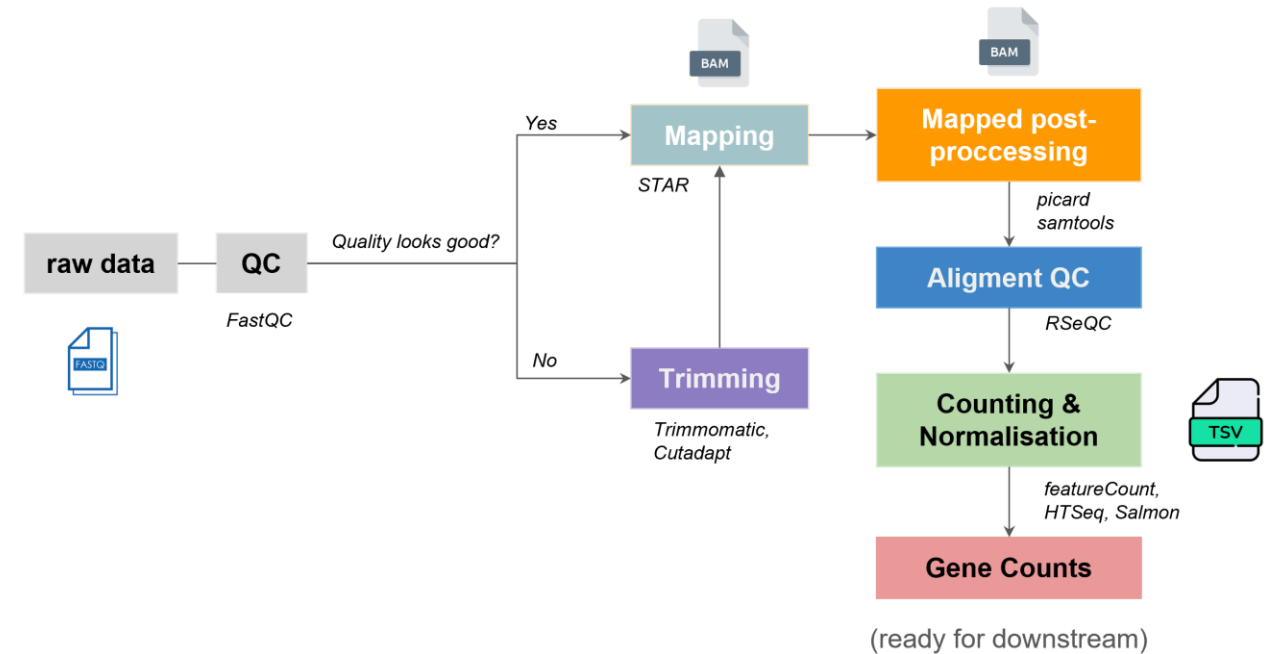


Upstream Analysis Workflow

Bulk DNA-seq

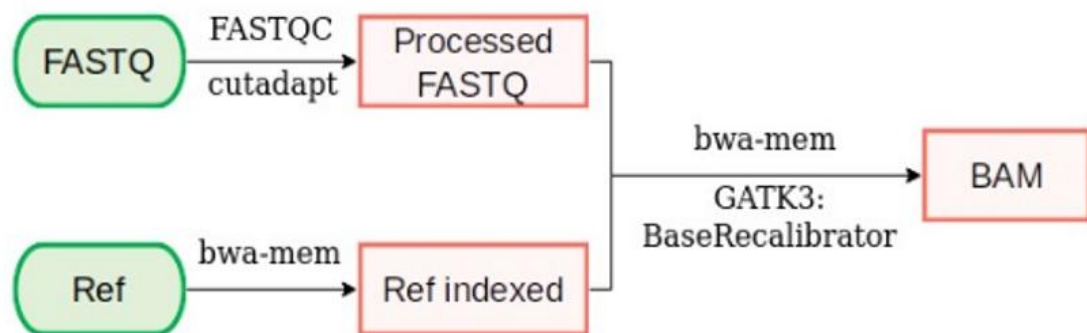


Bulk RNA-seq

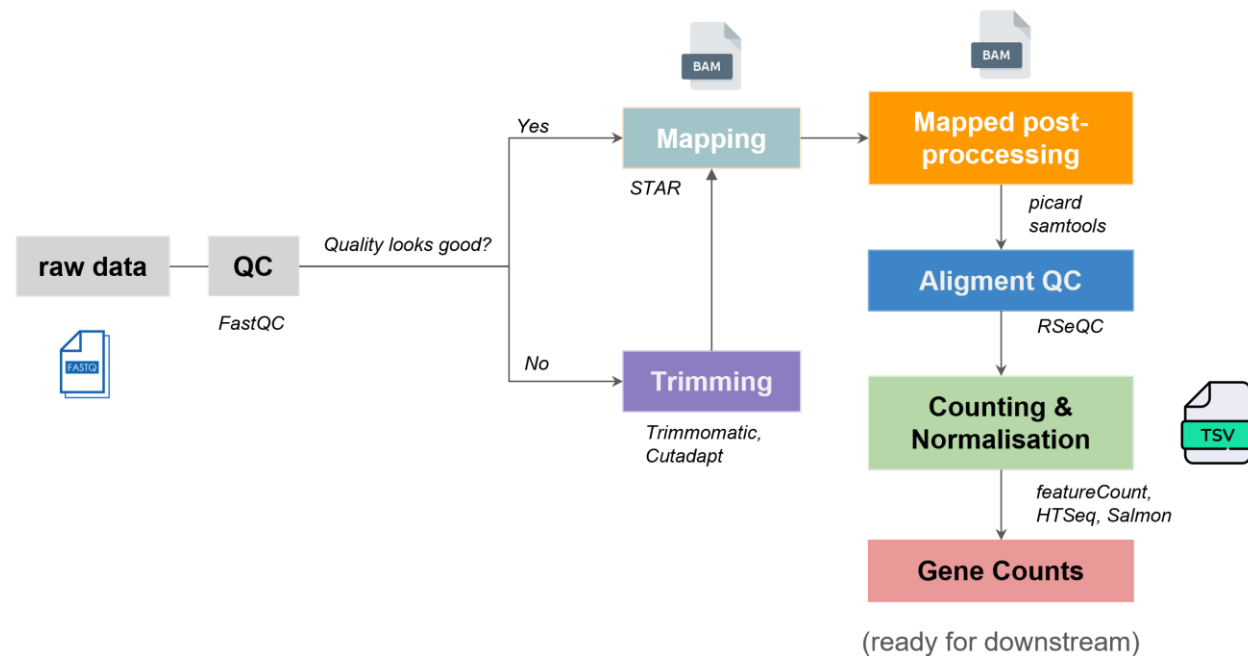


Upstream Analysis Workflow

Bulk DNA-seq



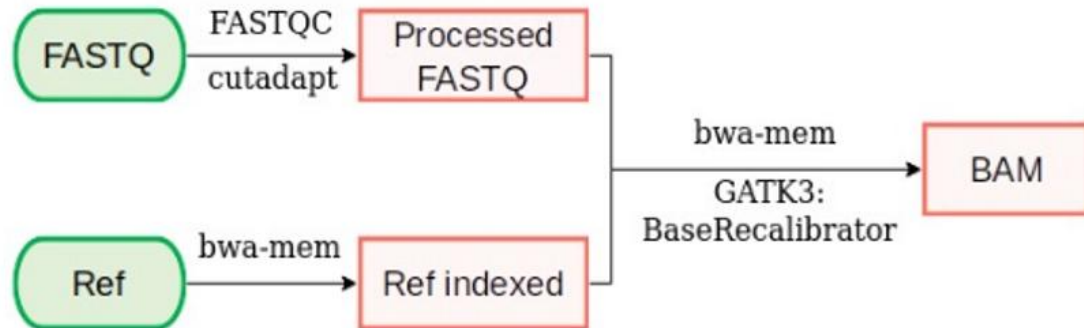
Bulk RNA-seq



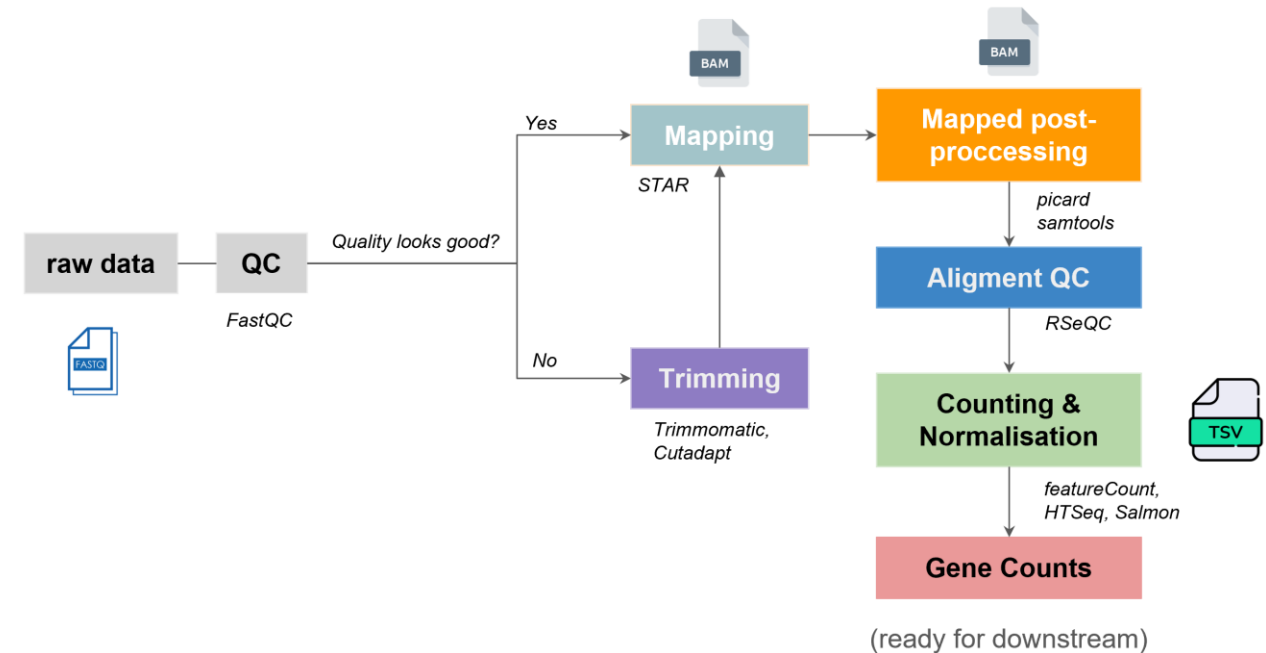
Where are the References need? And what are these?

Upstream Analysis Workflow

Bulk DNA-seq



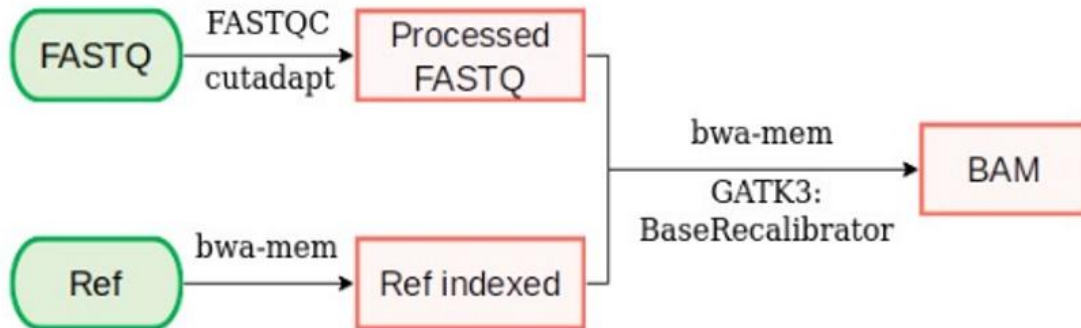
Bulk RNA-seq



Where are the References need? And what are these?

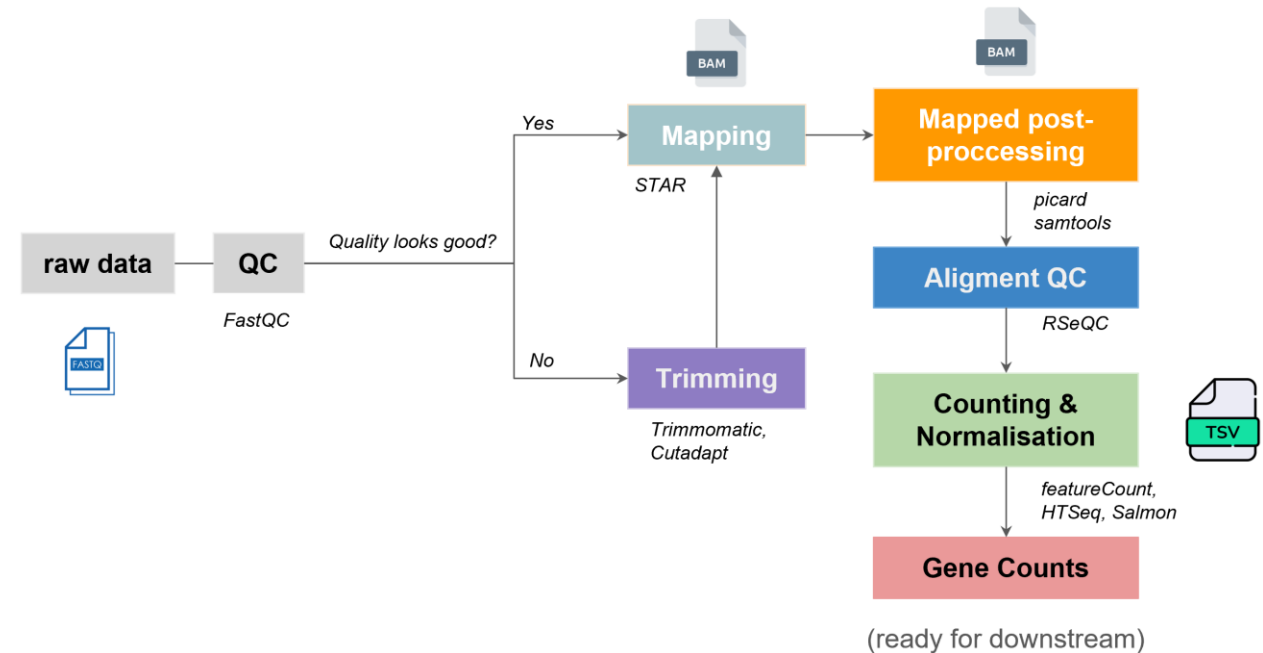
Upstream Analysis Workflow

Bulk DNA-seq



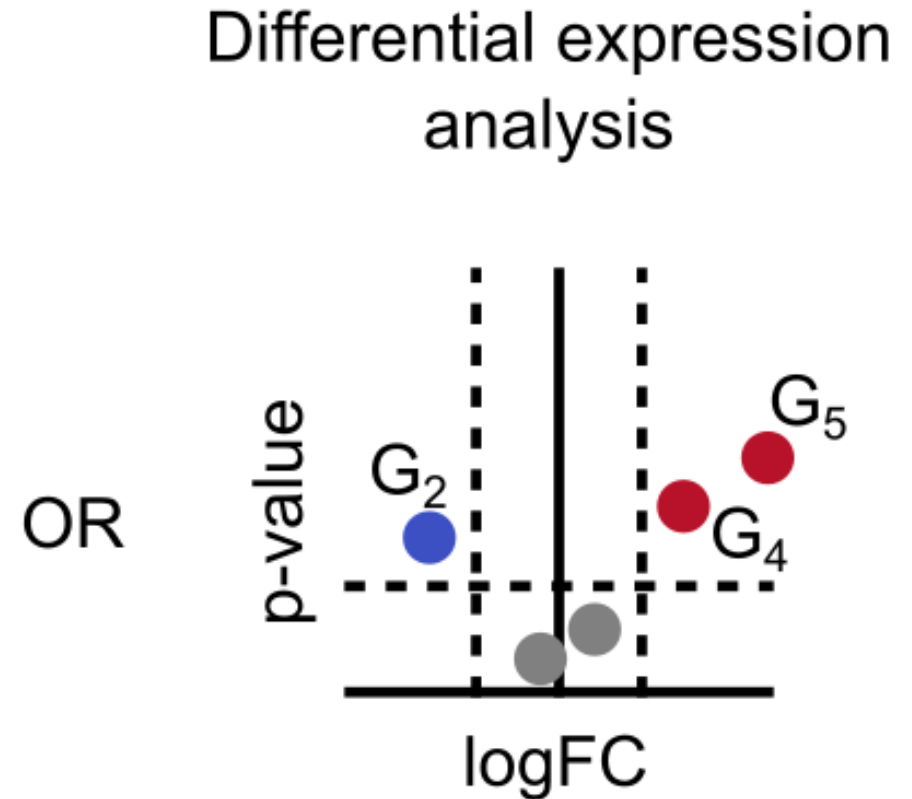
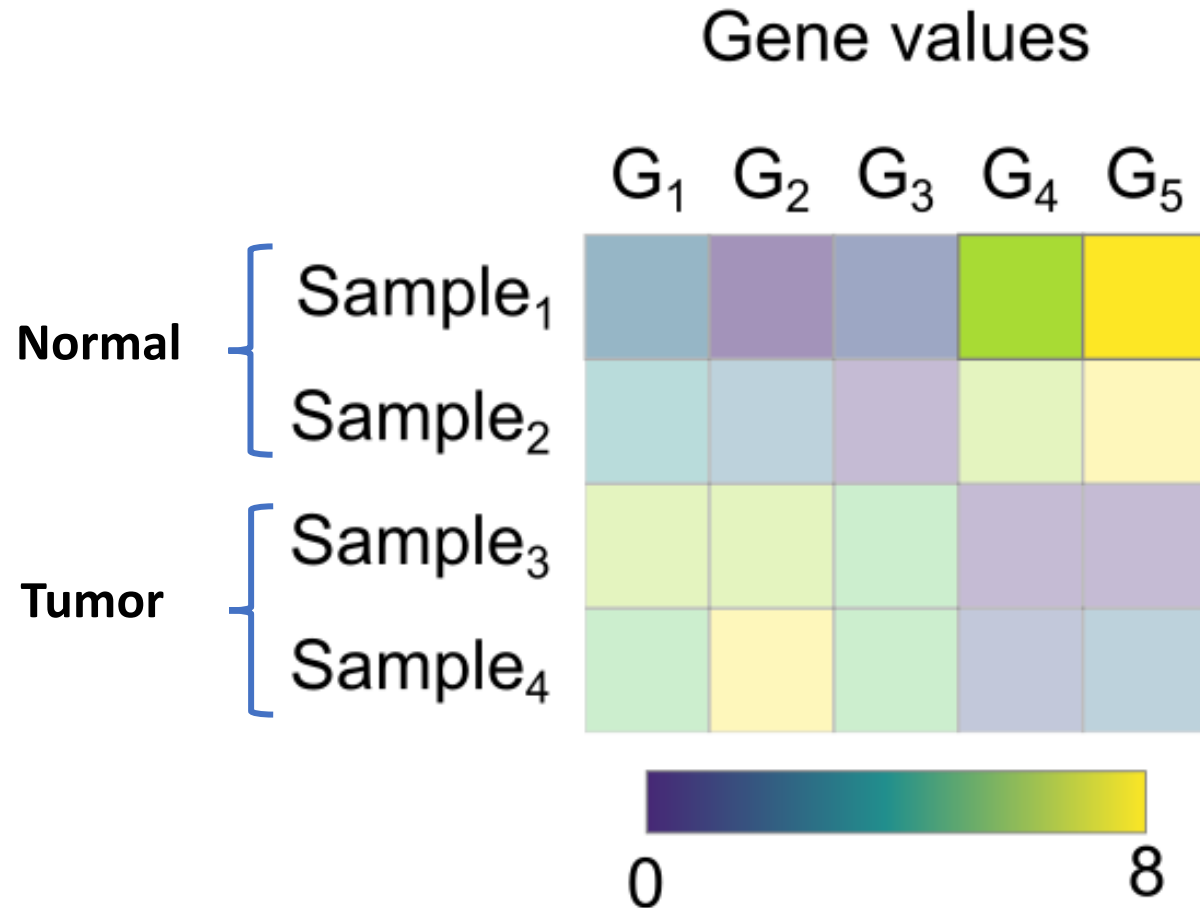
Human Genome Sequences can be found at UCSC, NCBI and Ensembl and GENCODE as fasta format

Bulk RNA-seq



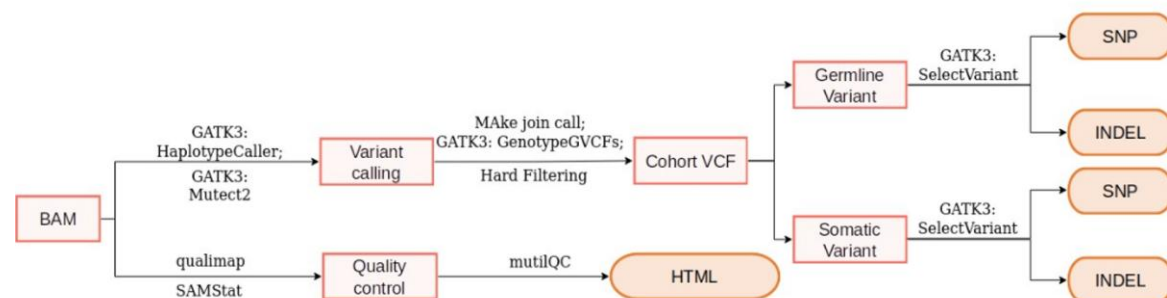
The four most common gene annotation databases are currently RefSeq, UCSC, Ensembl and GENCODE as GTF format

Differential Gene Expression

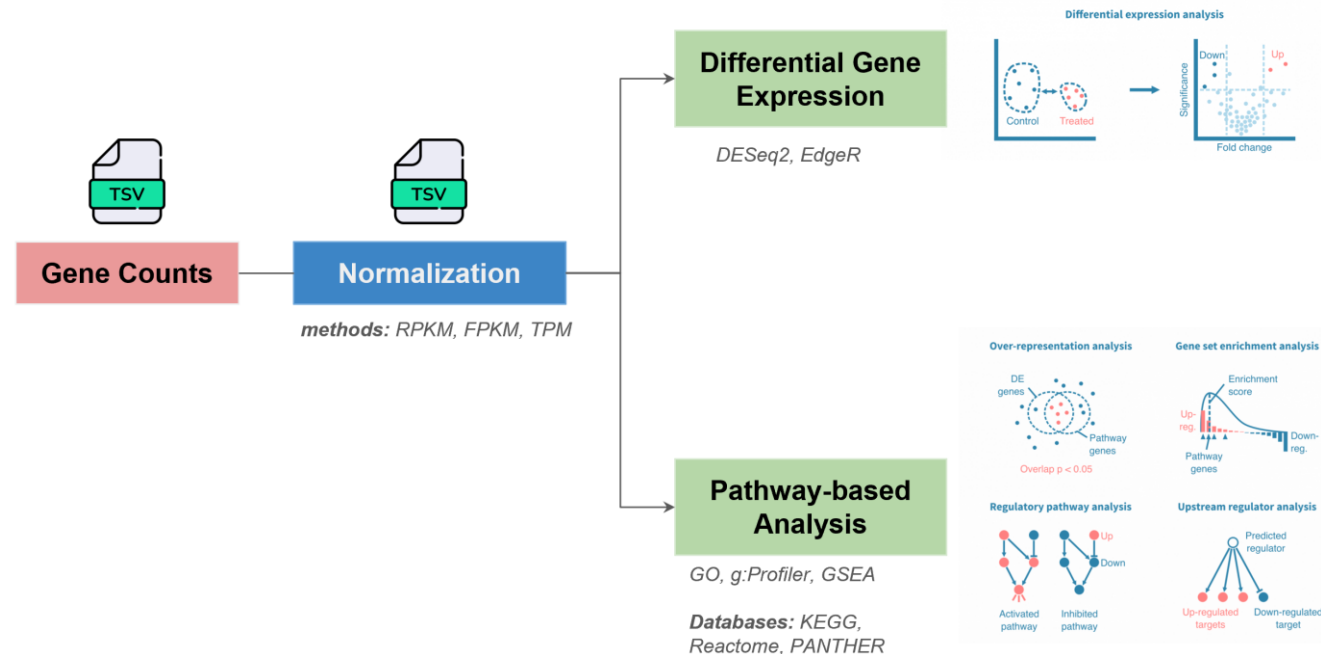


Downstream Analysis Workflow

Bulk DNA-seq

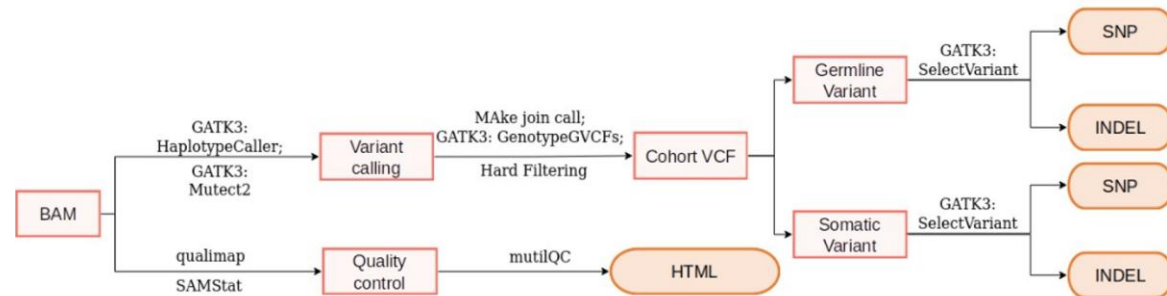


Bulk RNA-seq

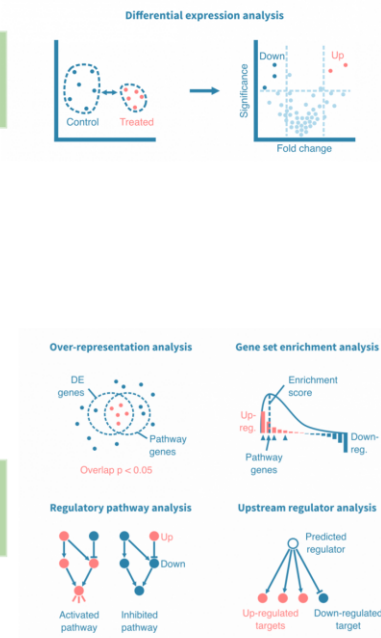
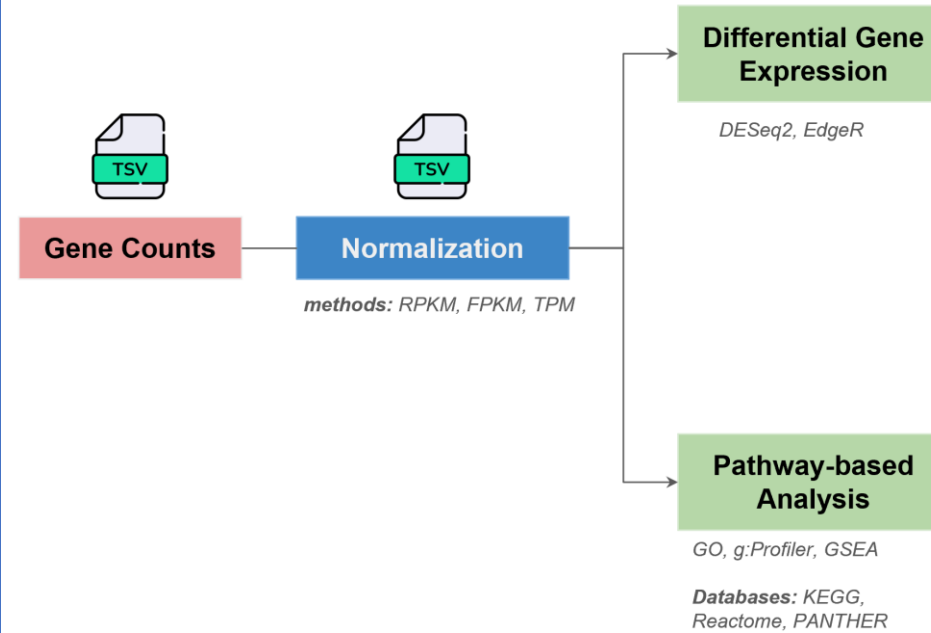


Downstream Analysis Workflow

Bulk DNA-seq



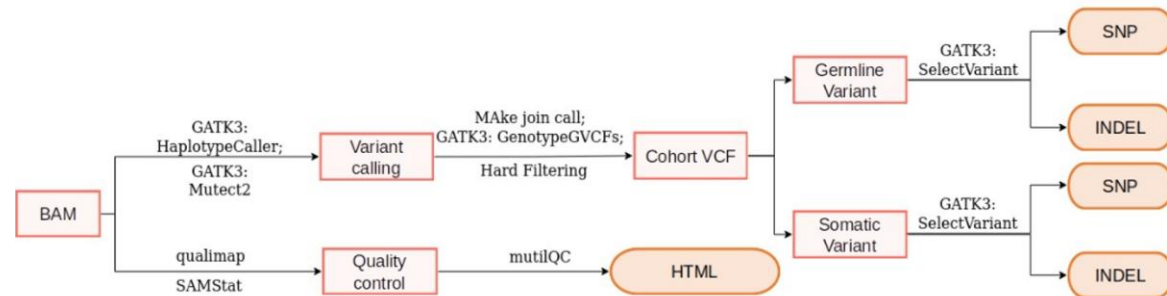
Bulk RNA-seq



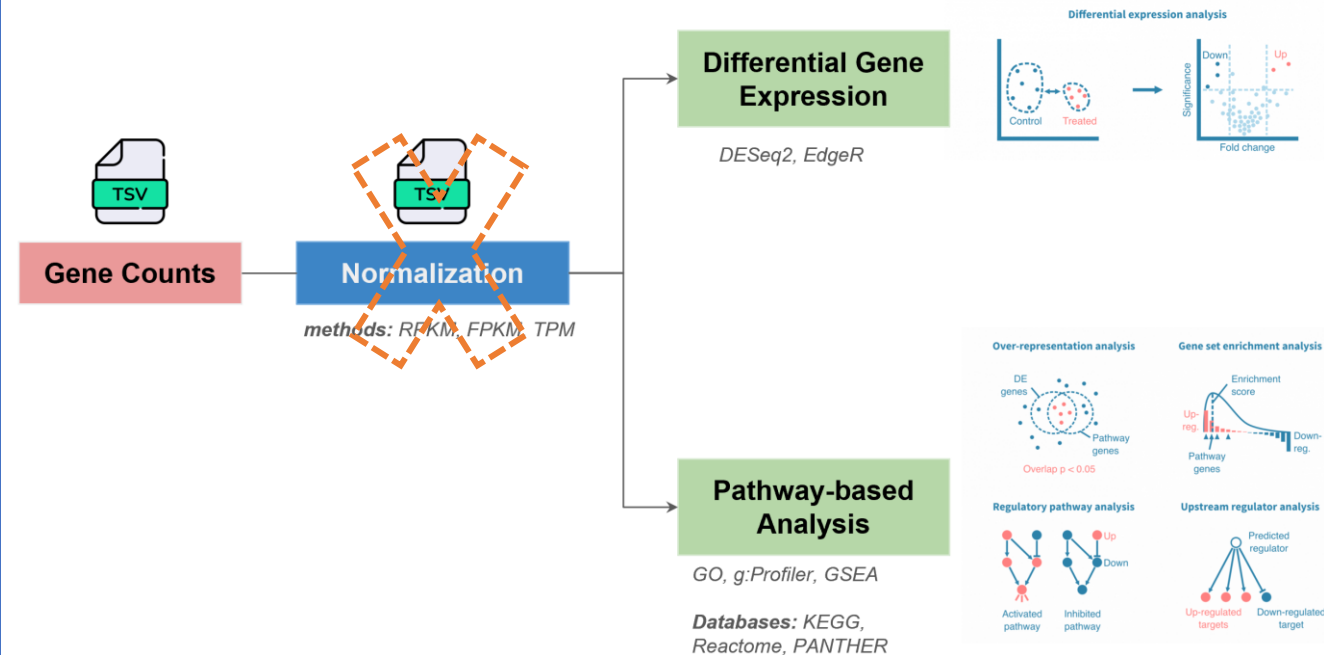
What is the unmatched?

Downstream Analysis Workflow

Bulk DNA-seq

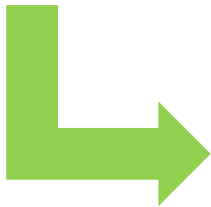


Bulk RNA-seq



What is the unmatched?

From BAM to Count



Gene	Sample 1 Healthy	Sample 2 Healthy	Sample 3 Healthy	Sample 4 Tumor	Sample 5 Tumor	Sample 6 Tumor
A	100	101	105	160	163	154
B	10	8	9	1	2	3
C	45	45	45	46	46	46
D	0.11	0.12	0.13	0.0012	0.0014	0.0013

Differential Gene Expression

Gene	Sample 1 Healthy	Sample 2 Healthy	Sample 3 Healthy	Sample 4 Tumor	Sample 5 Tumor	Sample 6 Tumor
A	100	101	105	160	163	154
B	10	8	9	1	2	3
C	45	44	45	46	47	46
D	0.11	0.12	0.13	0.0012	0.0014	0.0013

```
t.test(c(100, 101, 105), c(160, 163, 154))
```



Welch Two Sample t-test

data: c(100, 101, 105) and c(160, 163, 154)
t = -18.658, df = 3.2, p-value = 0.0002251
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
-66.38765 -47.61235
sample estimates:
mean of x mean of y
102 159

Gene A

```
t.test(c(0.11, 0.12, 0.13), c(0.0012, 0.0014, 0.0013))
```



Welch Two Sample t-test

data: c(0.11, 0.12, 0.13) and c(0.0012, 0.0014, 0.0013)
t = 20.558, df = 2.0004, p-value = 0.002355
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
0.09386214 0.14353786
sample estimates:
mean of x mean of y
0.1200 0.0013

Gene D

```
t.test(c(10, 8, 9), c(1, 2, 3))
```



Welch Two Sample t-test

data: c(10, 8, 9) and c(1, 2, 3)
t = 8.5732, df = 4, p-value = 0.001017
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
4.733042 9.266958
sample estimates:
mean of x mean of y
9 2

Gene B

```
t.test(c(45, 44, 45), c(46, 45, 46))
```



Welch Two Sample t-test

data: c(45, 44, 45) and c(46, 45, 46)
t = -2.1213, df = 4, p-value = 0.1012
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
-2.3088288 0.3088288
sample estimates:
mean of x mean of y
44.66667 45.66667

Gene C

Fisher's Exact Test: Example

Suppose we want to know whether or not gender is associated with political party preference. We take a simple random sample of 25 voters and survey them on their political party preference. The following table shows the results of the survey:

	Democrat	Republican	Total
Male	4	9	13
Female	8	4	12
Total	12	13	25

- H_0 : Gender and political party preference are independent.
- H_1 : Gender and political party preference are *not* independent.

Step 2: Calculated the two-tailed p value.

We can use the [Fisher's Exact Test Calculator](#) with the following input:

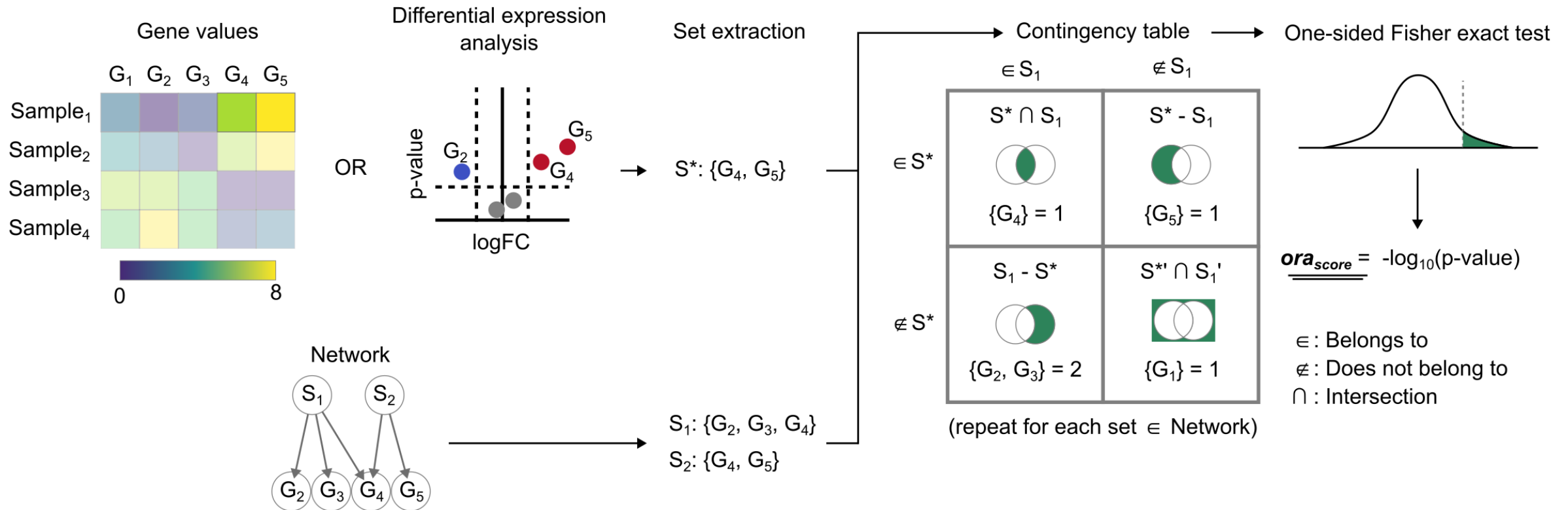
	Group 1	Group 2
Category 1	4	9
Category 2	8	4

CALCULATE

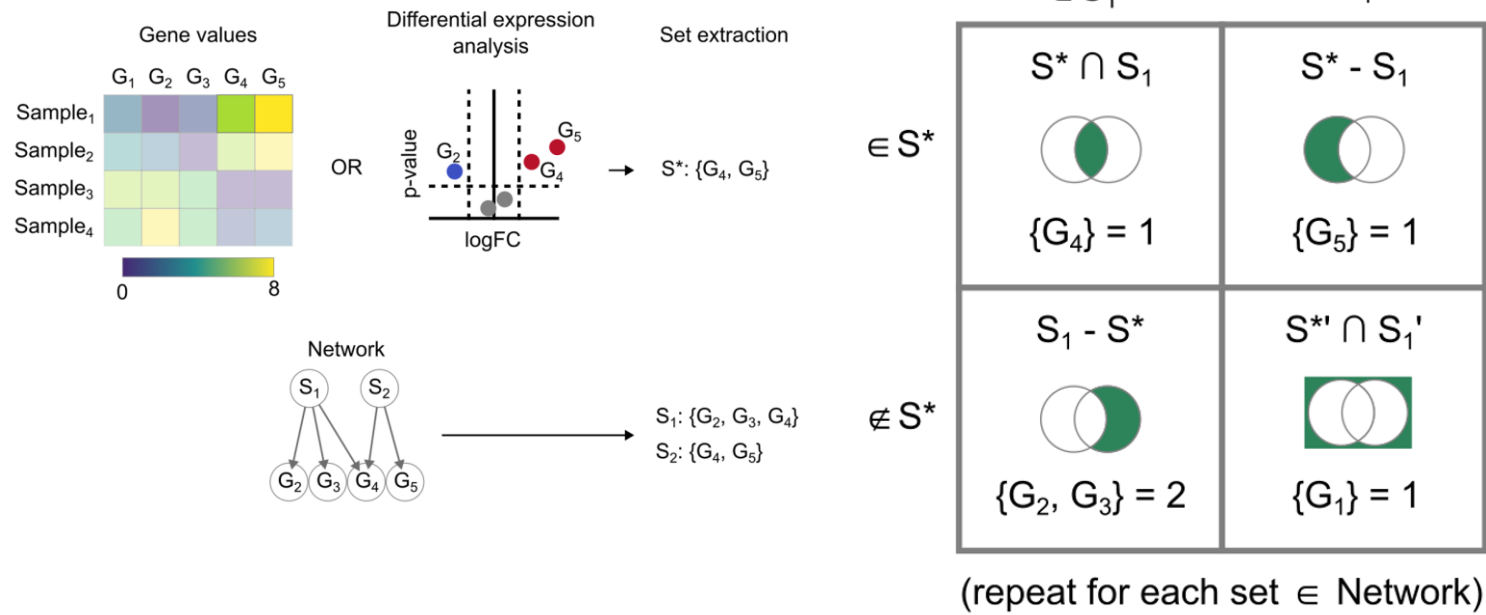
One-tailed p value: **0.081178**

Two-tailed p value: **0.115239**

Enrichment of DEGs with Over Representation Analysis (ORA)



Enrichment of DEGs with Over Representation Analysis (ORA)



$$\text{GeneRatio} = \frac{\text{Number of genes from your input list associated with the GO term}}{\text{Total number of genes in your input list}}$$

$$\text{BgRatio} = \frac{\text{Number of genes associated with the GO term in the background set}}{\text{Total number of genes in the background set}}$$

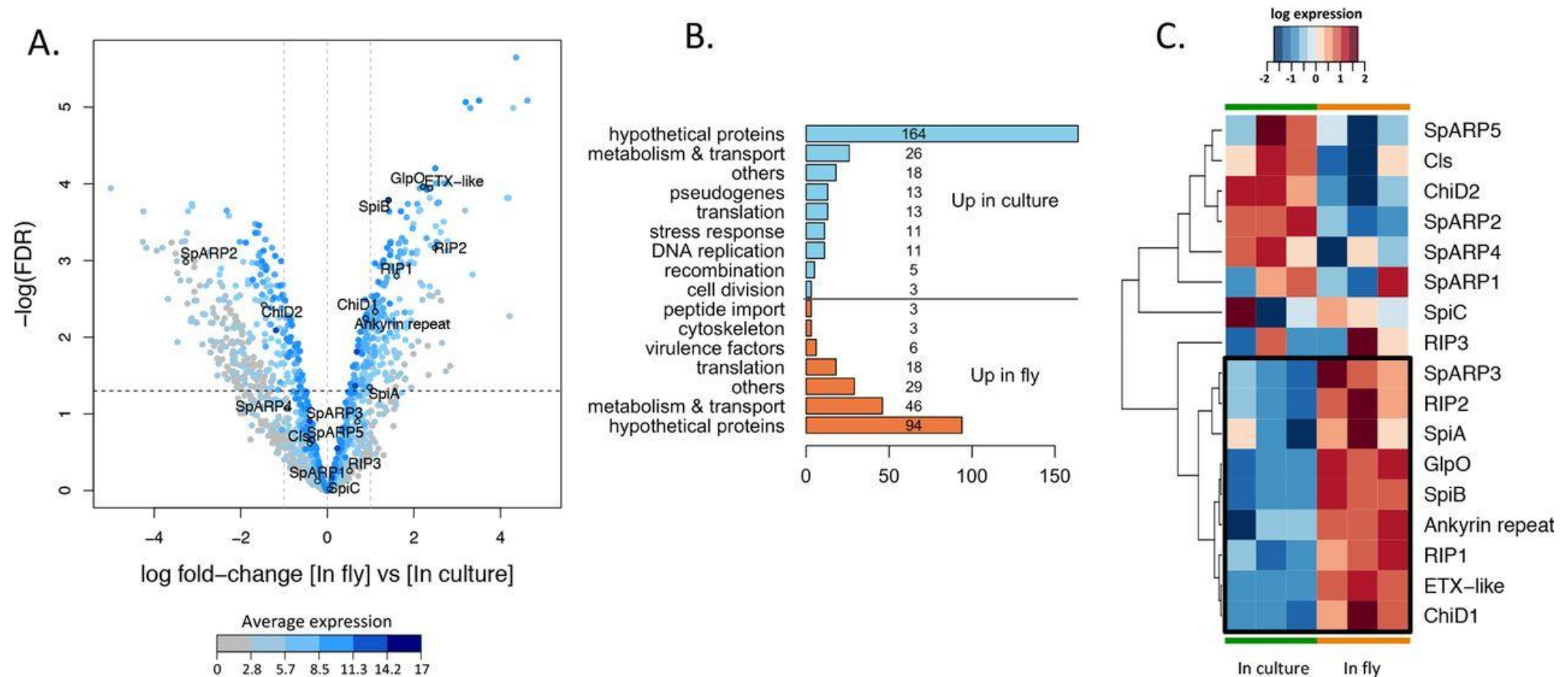
	ID	Description	GeneRatio	BgRatio
	<chr>	<chr>	<chr>	<chr>
GO:0016052	GO:0016052	carbohydrate catabolic process	23/238	112/6476
GO:0044282	GO:0044282	small molecule catabolic process	25/238	145/6476
GO:0005975	GO:0005975	carbohydrate metabolic process	34/238	277/6476
GO:0032787	GO:0032787	monocarboxylic acid metabolic process	25/238	170/6476
GO:0006091	GO:0006091	generation of precursor metabolites and energy	25/238	217/6476
GO:0006979	GO:0006979	response to oxidative stress	16/238	103/6476

Enrichment of DEGs with Over Representation Analysis (ORA)

	ID	Description	GeneRatio	BgRatio	RichFactor	FoldEnrichment	zScore	pvalue	p.adjust	qvalue
	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
GO:0016052	GO:0016052	carbohydrate catabolic process	23/238	112/6476	0.2053571	5.587785	9.566051	8.859134e-12	6.856970e-09	5.856354e-09
GO:0044282	GO:0044282	small molecule catabolic process	25/238	145/6476	0.1724138	4.691394	8.780594	6.069770e-11	2.349001e-08	2.006219e-08
GO:0005975	GO:0005975	carbohydrate metabolic process	34/238	277/6476	0.1227437	3.339866	7.774200	3.099557e-10	7.996858e-08	6.829902e-08
GO:0032787	GO:0032787	monocarboxylic acid metabolic process	25/238	170/6476	0.1470588	4.001483	7.745849	2.010853e-09	3.891001e-07	3.323199e-07
GO:0006091	GO:0006091	generation of precursor metabolites and energy	25/238	217/6476	0.1152074	3.134802	6.247708	2.930343e-07	4.536170e-05	3.874221e-05
GO:0006979	GO:0006979	response to oxidative stress	16/238	103/6476	0.1553398	4.226809	6.447712	8.643298e-07	9.716547e-05	8.298642e-05

The qvalue in the results table from enrichGO represents the adjusted p-value for multiple testing correction. It is typically calculated using the False Discovery Rate (FDR) method, which controls the proportion of false positives among the list of enriched terms.

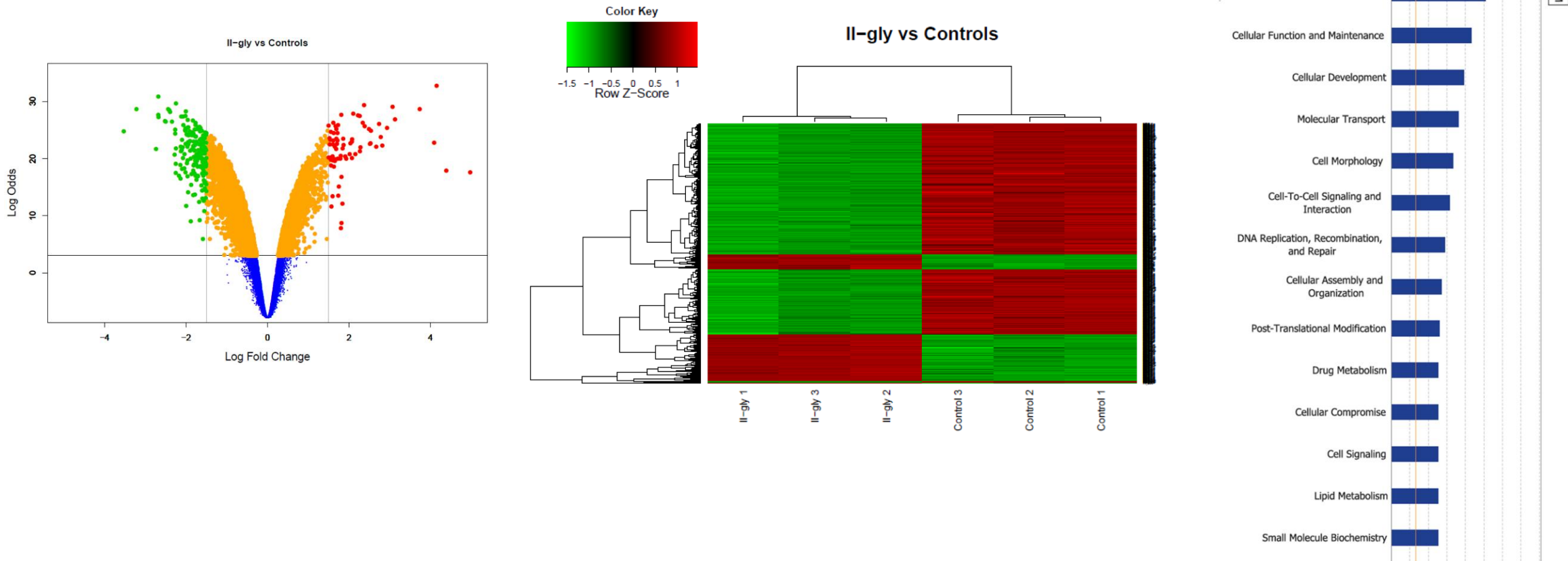
Exercise 1: Explain the below figure



Exercise 1: Explain the below figure

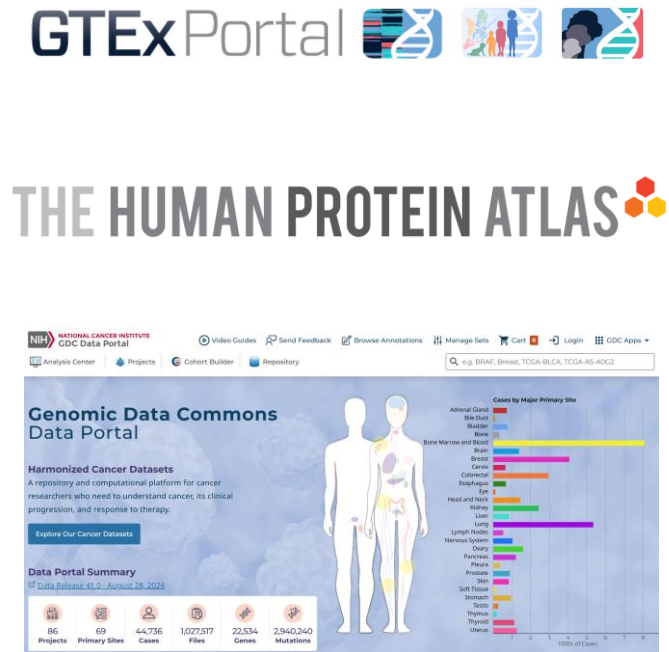
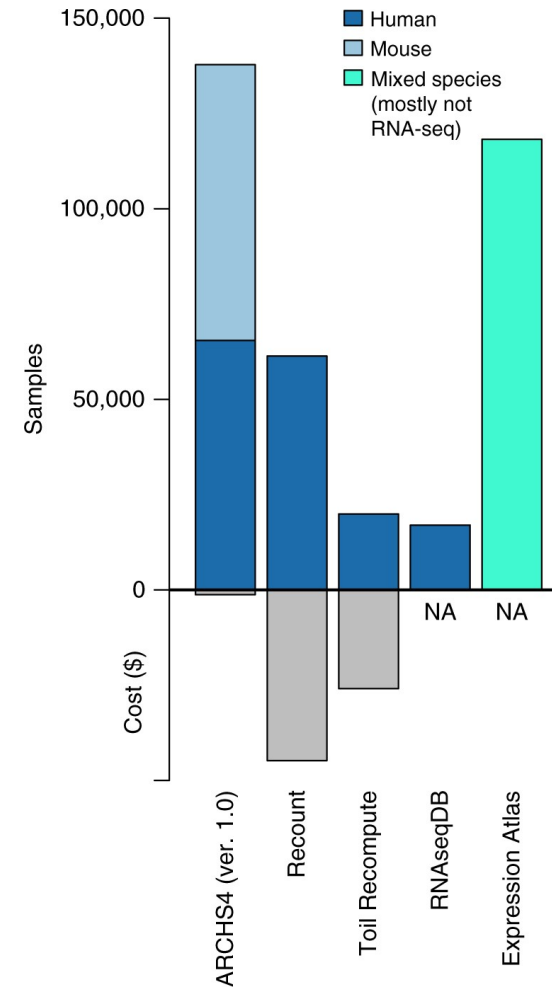
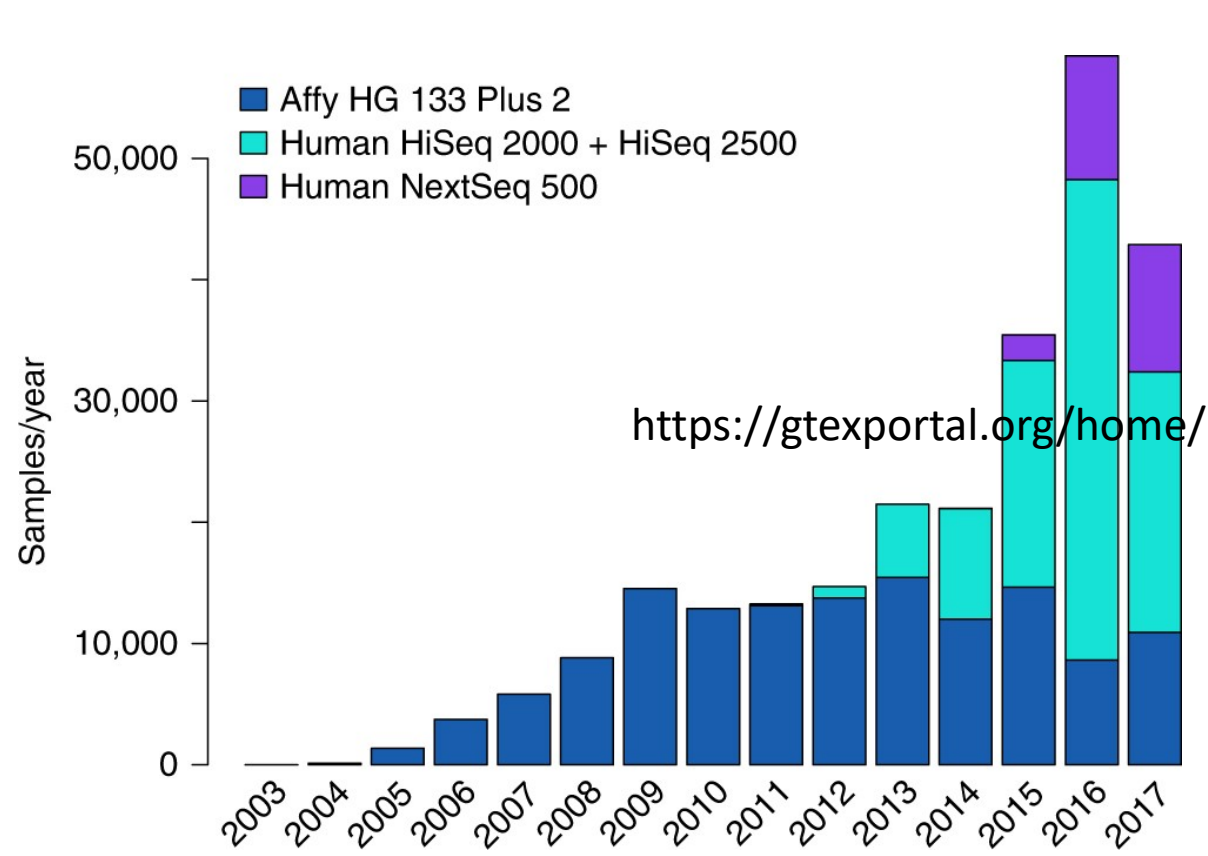
FIG 3 (A) Volcano plot of differential gene expression of *S. poulsonii* in host versus in culture. Each point represents the average value of one transcript in three replicate experiments. The expression difference is considered significant for a log₂ fold change of ≥ 1 (outer light gray broken vertical lines) and for a P value of ≤ 0.05 [$-\log(\text{FDR})$ of ≥ 1.3 , dark broken horizontal line]. Points are colored according to their average expression in all data sets. Names and outlined points represent virulence factors. FDR, false-discovery rate. (B) Manual clustering of the transcripts differentially expressed by *S. poulsonii* in the fly versus in the culture. The numbers of sequences in the different categories are indicated on the bars or to the right of the bars. (C) Heatmap of *S. poulsonii* virulence gene expression. Each column represents the value for one replicate experiment in culture or in the fly. The colors represent the log₁₀ level of expression in the corresponding experiment. The cluster of genes that are induced when *S. poulsonii* is in the host (versus in vitro) is shown enclosed in a black box. SpARP5, *S. poulsonii* ARP5.

Exercise 2: Explain the below figure

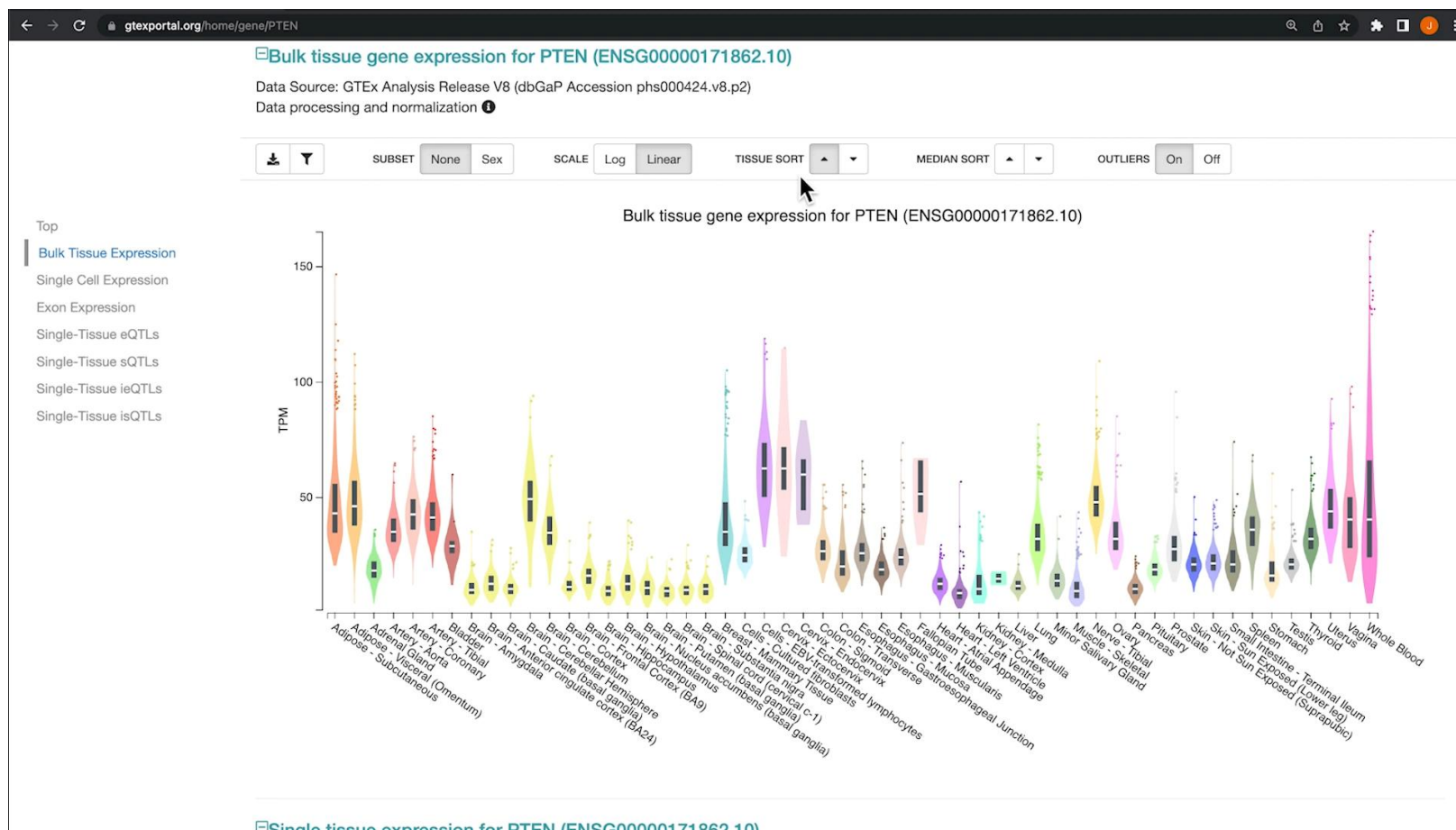


<https://doi.org/10.1371/journal.pone.0054664>

RNA-seq databases: GTEx and TCGA



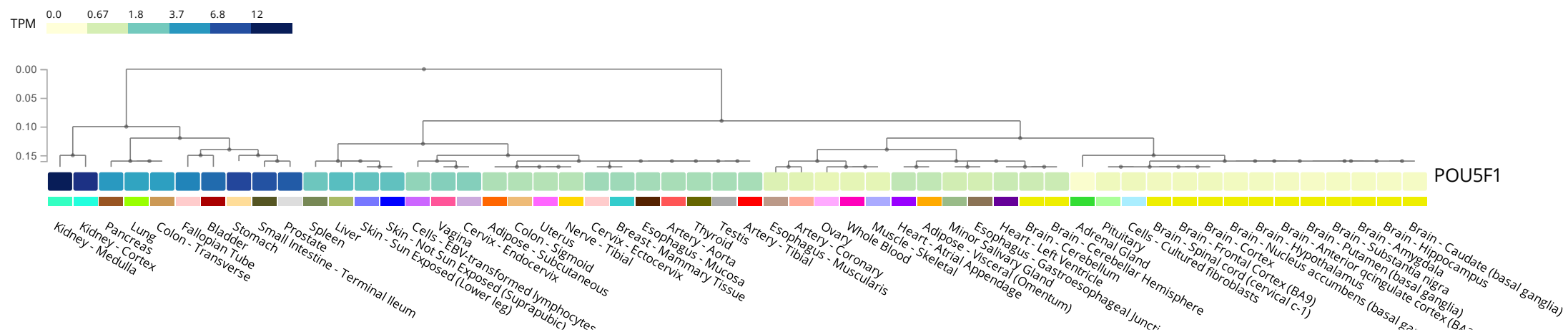
GTEx Portal

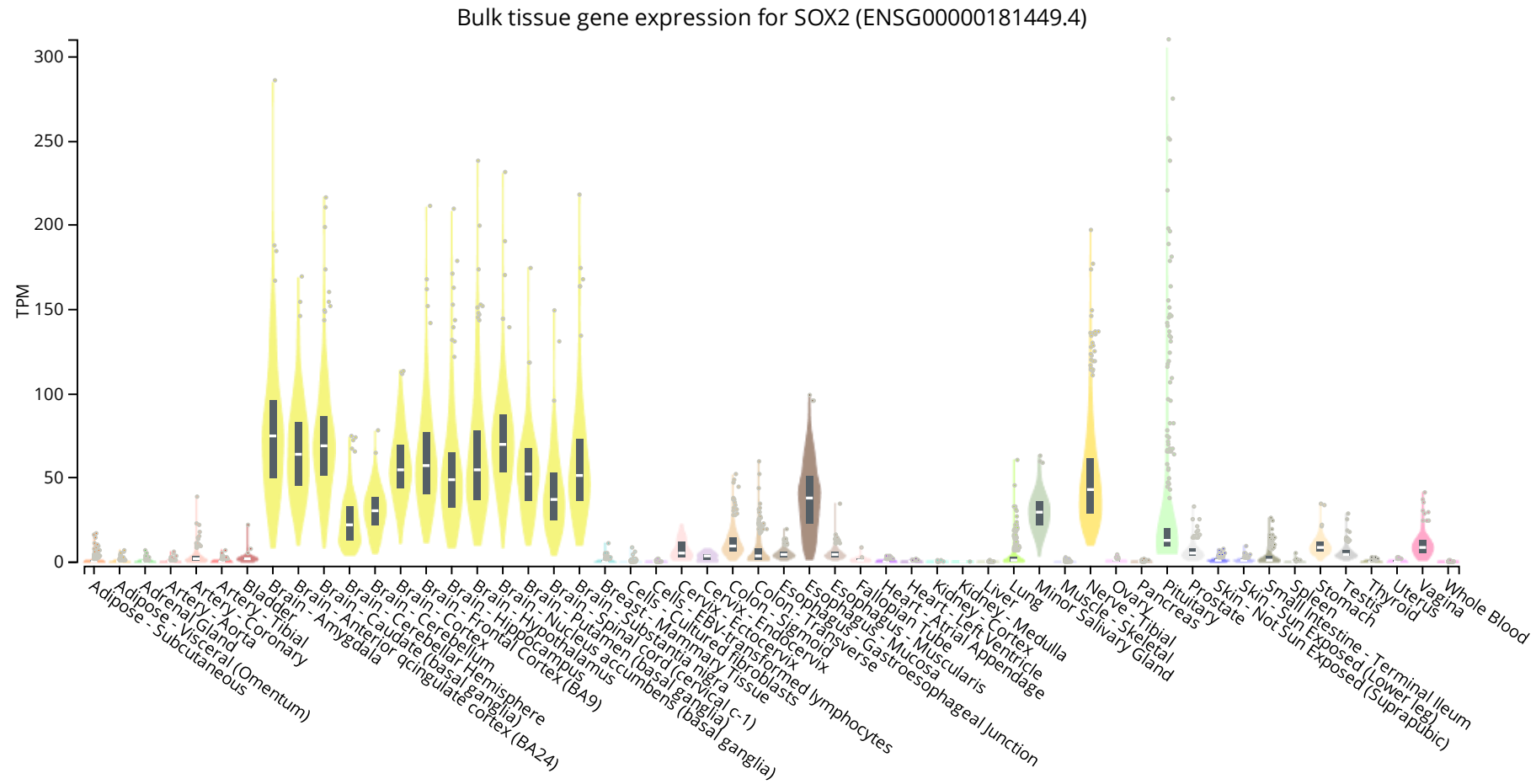


https://www.youtube.com/watch?v=WjHkb_y_yFk&t=2s

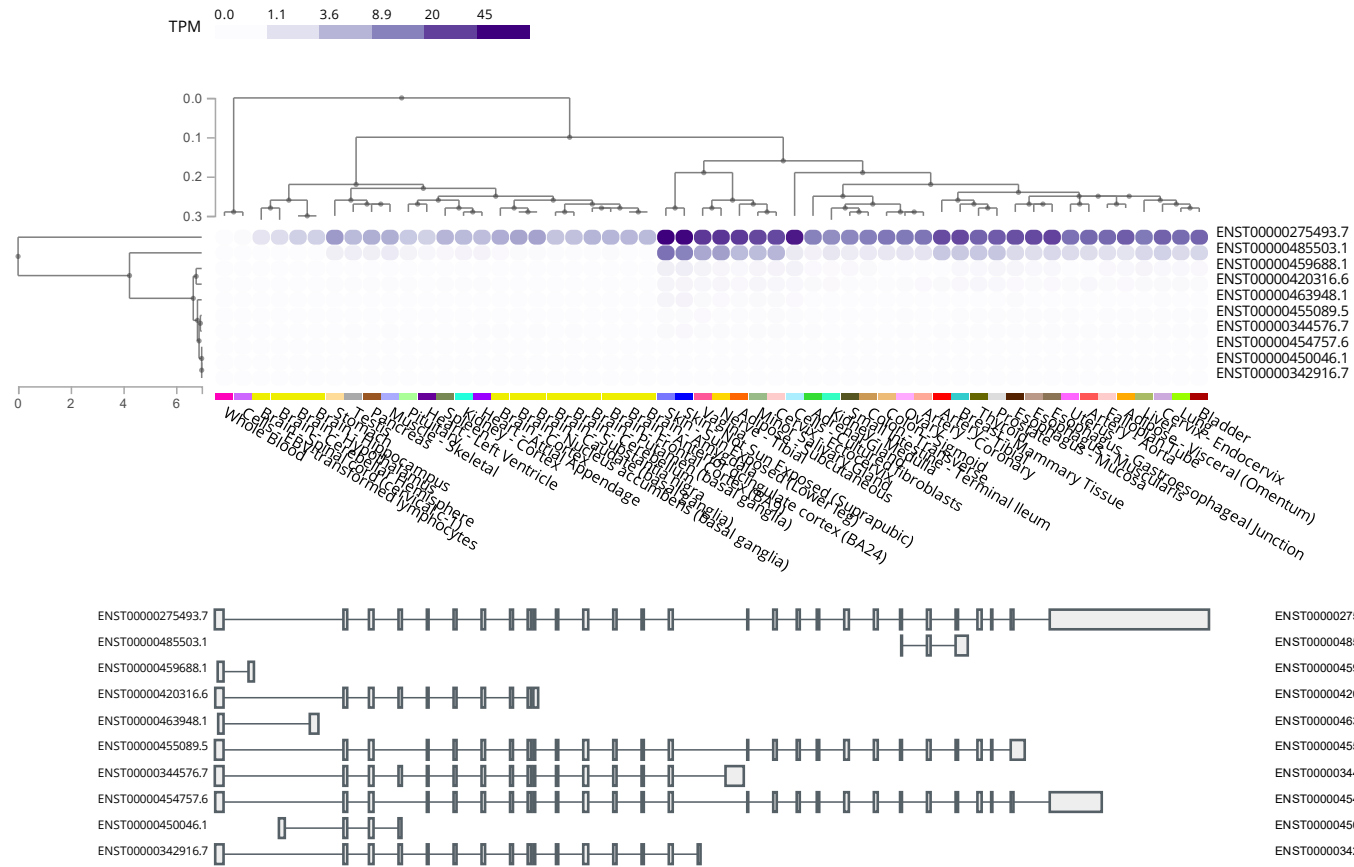
<https://gtexpportal.org/home/>

GTEx Portal





Isoform Expression of EGFR: ENSG00000146648.20 epidermal growth factor receptor [Source:HGNC Symbol;Acc:HGNC:3236]



Genomic Data Commons Data Portal

Harmonized Cancer Datasets

A repository and computational platform for cancer researchers who need to understand cancer, its clinical progression, and response to therapy.

Explore Our Cancer Datasets

Data Portal Summary

[Data Release 42.0 - January 30, 2025](#)



86
Projects



69
Primary Sites



44,736
Cases



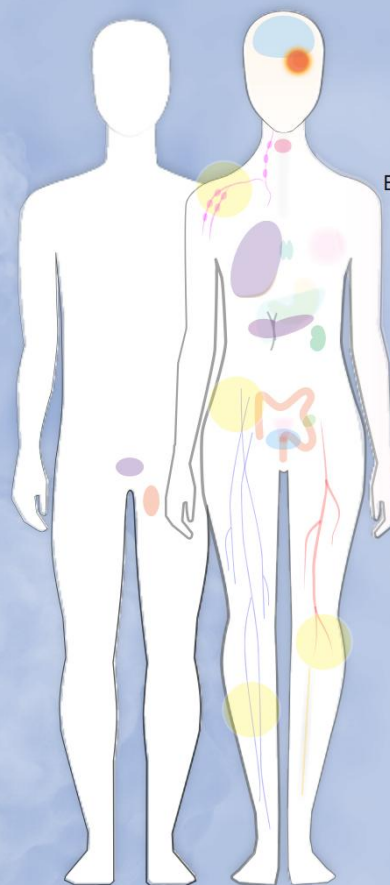
1,121,816
Files



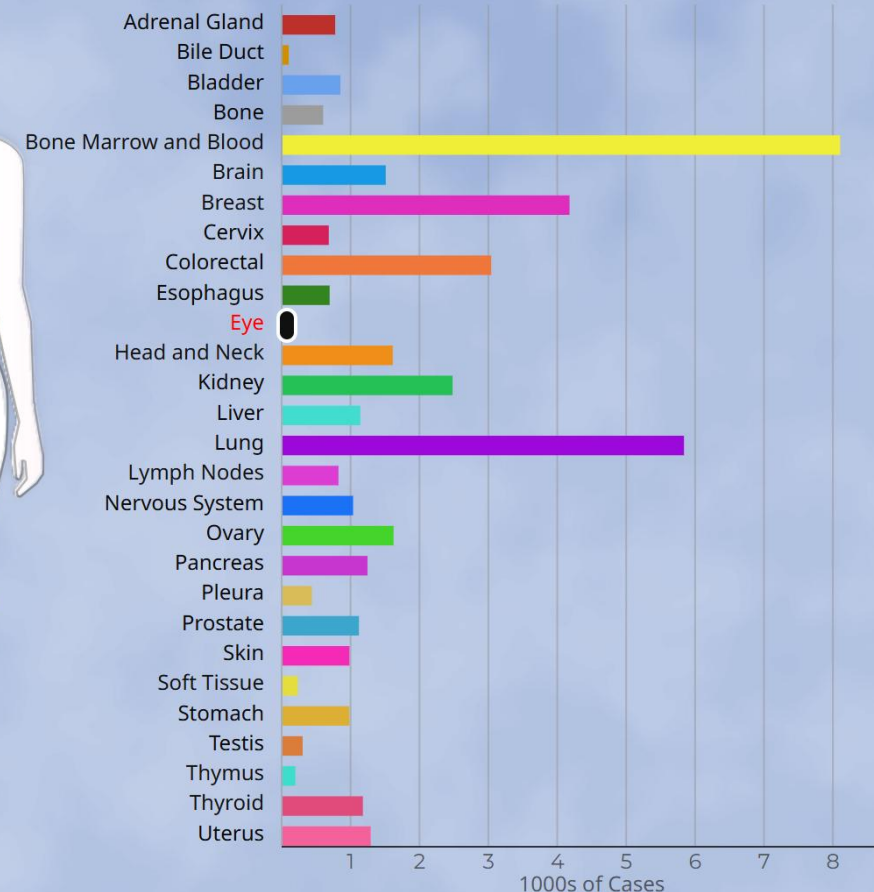
22,534
Genes



2,940,240
Mutations



Cases by Major Primary Site



ANALYSIS TOOLS



BAM Slicing Download ▾

80 Cases



Clinical Data Analysis ▾

114 Cases



Cohort Comparison ▾

114 Cases



OncoMatrix ▾

80 Cases



ProteinPaint ▾

80 Cases



Single Cell RNA-seq ▾

0 Cases ⓘ



Cohort Level MAF ▾

80 Cases



Gene Expression Clustering ▾

80 Cases



Mutation Frequency ▾

80 Cases

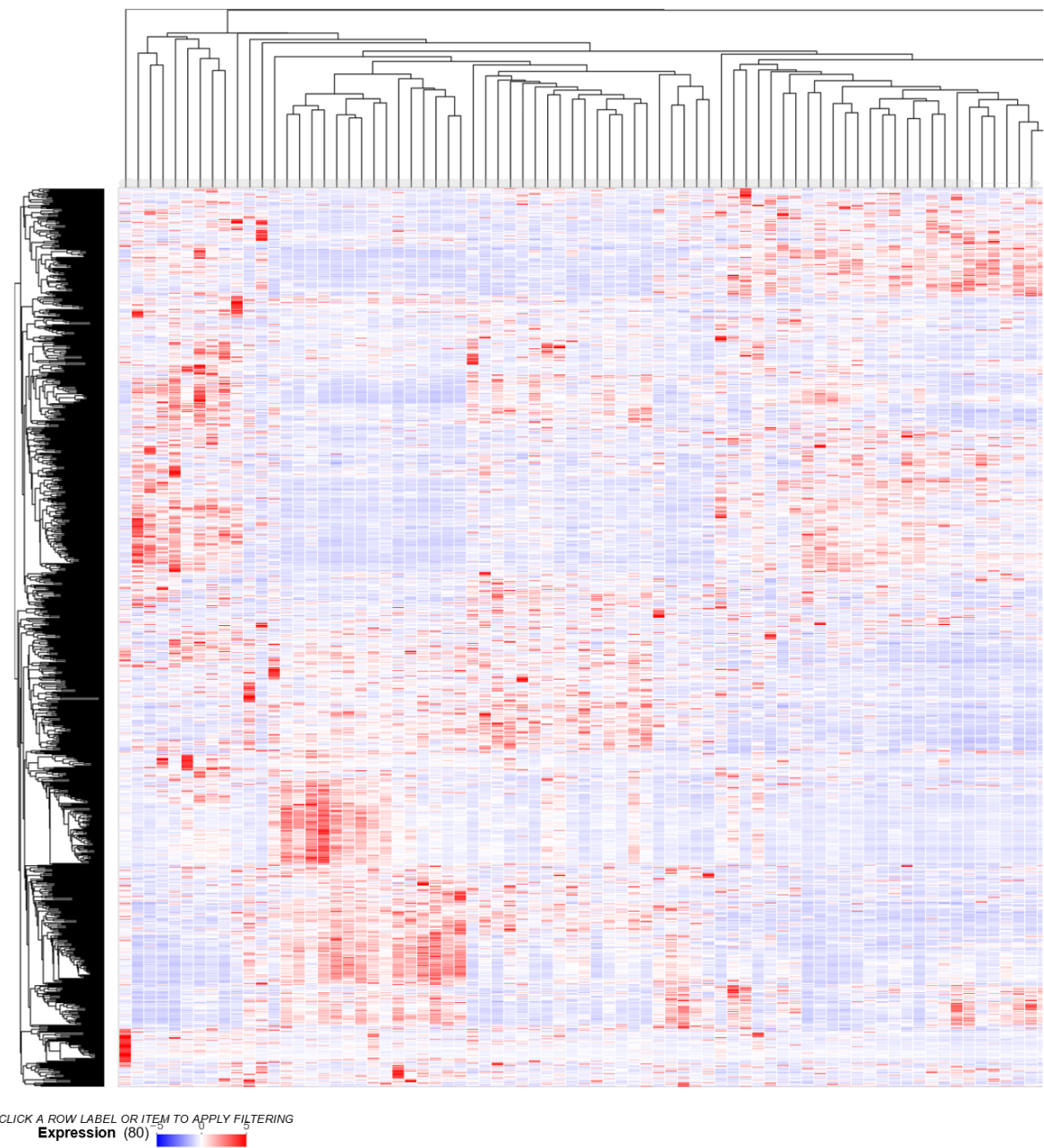


Sequence Reads ▾

80 Cases



Set Operations ▾



Cảm ơn các thầy cô và các bạn!

Ngày 29 tháng 03 năm 2025

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Zalo: 0901802182